

Bash Scripting & Automation

The Security Professional's Glue

Bash is how security professionals glue tools together: "grab those IPs, check them here, and alert me."

Variables, Conditionals, Loops

```
#!/bin/bash
TARGET="192.168.1.0/24"
for ip in $(seq 1 254); do
    ping -c 1 -W 1 192.168.1.$ip > /dev/null 2>&1 && echo "192.168.1.$ip is up"
done

if [ -f /var/log/auth.log ]; then
    grep "Failed" /var/log/auth.log | wc -l
else
    echo "missing auth.log"
fi
```

Parsing Logs Like a Boss

```
# Top source IPs in failed logins
grep "Failed password" auth.log | awk '{print $(NF-3)}' | sort | uniq -c | sort -rn | head

# Would you block these IPs? Extract them:
grep "Failed password" auth.log | awk '{print $(NF-3)}' | sort | uniq | wc -l
```

Automation Patterns for Blue Team

```
#!/bin/bash
# daily recon baseline
OUT=~/.baseline-$(date +%F).txt
ss -tulnp > $OUT
find / -perm -4000 2>/dev/null >> $OUT
```

```
ls -la /var/mail >> $OUT
echo "Baseline saved: $OUT"
```

Automation Patterns for Red Team

```
#!/bin/bash
# quick DNS sweep (change example.com)
for sub in www admin dev prod mail ftp vpn old; do
    host $sub.example.com 2>/dev/null | grep "has address" && echo "FOUND: $sub"
done
```

Functions & Error Handling

```
check_port() {
    nc -zv -w 2 "$1" "$2" 2>&1 | grep -q succeeded && echo "OPEN $1:$2"
}
check_port 10.0.0.5 22
check_port 10.0.0.5 445
```

Professional habit: every script you write should (1) work from anywhere, (2) not depend on interactive prompts, (3) print what it's doing, and (4) log or save output. Your future SOC will run your scripts unattended.

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